

Periodic Table

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Mendeleev's Periodic Law

The physical and chemical properties of elements are periodic functions of their atomic weights.

Modern Periodic Law:

The physical and chemical properties of elements are periodic functions of their atomic numbers.

- The modern long form of periodic table was constructed based on above law. The following points are considered while constructing the periodic table.
- The elements in the periodic table are arranged in the increasing order of the atomic number.
- Every row, also called as period, in the periodic table starts with the filling up of differentiating electron into a new quantum shell.
- The elements in a vertical column called as group should get similar outer electronic configuration since it is observed that the elements with similar outer electronic configuration show similar chemical properties.

Periodic Table of the Elements

1 1IA 1A	2 IIA 2A											13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 H Hydrogen 1.0079																	2 He Helium 4.00260
3 Li Lithium 6.941	4 Be Beryllium 9.01218											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.00644	8 O Oxygen 15.9994	9 F Fluorine 18.998403	10 Ne Neon 20.1797
11 Na Sodium 22.989769	12 Mg Magnesium 24.305	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 Al Aluminum 26.981539	14 Si Silicon 28.0855	15 P Phosphorus 30.973762	16 S Sulfur 32.066	17 Cl Chlorine 35.4527	18 Ar Argon 39.948
19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.95591	22 Ti Titanium 47.88	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938	26 Fe Iron 55.847	27 Co Cobalt 58.9332	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.39	31 Ga Gallium 69.723	32 Ge Germanium 72.64	33 As Arsenic 74.92159	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.80
37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90585	40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.94	43 Tc Technetium 98.9072	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.9055	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.71	51 Sb Antimony 121.760	52 Te Tellurium 127.5	53 I Iodine 126.90447	54 Xe Xenon 131.29
55 Cs Cesium 132.90543	56 Ba Barium 137.327	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.9479	74 W Tungsten 183.85	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.9665	80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98037	84 Po Polonium [209]	85 At Astatine 208.9871	86 Rn Radon 222.0176
87 Fr Francium 223.0187	88 Ra Radium 226.0254	89-103	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [269]	111 Rg Roentgenium [272]	112 Cn Copernicium [277]	113 Uut Ununtrium unknown	114 Uuq Ununquadium [289]	115 Uup Ununpentium unknown	116 Uuh Ununhexium [298]	117 Uus Ununseptium unknown	118 Uuo Ununoctium unknown
Lanthanide Series		57 La Lanthanum 138.9055	58 Ce Cerium 140.115	59 Pr Praseodymium 140.90765	60 Nd Neodymium 144.24	61 Pm Promethium 144.9127	62 Sm Samarium 150.36	63 Eu Europium 151.9655	64 Gd Gadolinium 157.25	65 Tb Terbium 158.92534	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93032	68 Er Erbium 167.26	69 Tm Thulium 168.93421	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.967	
Actinide Series		89 Ac Actinium 207.03312	90 Th Thorium 232.0377	91 Pa Protactinium 231.03688	92 U Uranium 238.02891	93 Np Neptunium 237.04817	94 Pu Plutonium 244.0642	95 Am Americium 243.0613	96 Cm Curium 247.0713	97 Bk Berkelium 247.0713	98 Cf Californium 251.0788	99 Es Einsteinium [252]	100 Fm Fermium [257]	101 Md Mendelevium [258]	102 No Nobelium [259]	103 Lr Lawrencium [260]	
		Alkali Metal	Alkaline Earth	Transition Metal	Basic Metal	Semimetals	Nonmetals	Halogens	Noble Gas	Lanthanides	Actinides						

Properties of periodic table

- Each period starts with an alkali metal and ends with an inert gas element.
- The **first period** is a very short period with only two elements i.e., Hydrogen (H) & Helium (He). In this period, the 1s orbital is being filled up.
- The **second period** starts with Lithium (Li) and ends with Neon (Ne) and contains 8 elements. It is called first short period. In this group, the 2s & 2p orbitals are being filled up.
- The **third period** also contain 8 elements i.e., from Sodium (Na) to Argon (Ar). It is called second short period. The 3s & 3p orbitals are being filled up in this period.
- The **fourth period** is the first long period with 18 elements, it starts with Potassium (K) and ends with Krypton (Kr). It also includes 10 elements belonging to 3d series i.e., from Scandium (Sc) to Zinc (Zn). In this period, not only 4s & 4p and also the 3d orbitals are being filled up by electrons.

- The **fifth period** is the second long period with 18 elements, it starts with Rubidium (Rb) and ends with Xenon (Xe). It also includes 10 elements belonging to 4d series i.e. from Yttrium (Y) to Cadmium (Cd). The 5s & 5p along with 4d orbitals are filled up by electrons.
- The **sixth period** is the longest period with 32 elements. It not only includes 10 elements belonging to 5d series i.e., Lanthanum (La), Hafnium (Hf) to Mercury (Hg) but also contains 14 elements belonging the 4f series called lanthanides (Cerium (Ce) to Lutetium (Lu)).

In this period, the 6s & 6p along with 4f & 5d orbitals are being filled up.

- The **seventh period** is an incomplete period. It starts with Fr. It also includes the 14 elements belonging to 5f series called actinides (Thorium (Th) to Lawrencium (Lr)).

In this period, the 7s & 5f orbitals are filled up. Until recently this period is considered to be incomplete. The names of some of its elements are not yet finalized (like Uut, Uup etc.)

- The **Lanthanides** and **actinides** are placed below the periodic table separately.

Groups:

- The 18 groups in the periodic table are numbered from 1 to 18 according to IUPAC convention.
- The elements of IA to VII A groups i.e., 1, 2, 13, 14, 15, 16 & 17 groups are called as **representative elements**.
- The zero group (or 18th group) elements are called as **inert gases** or **noble gases**. This group includes He, Ne, Ar, Kr, Xe and Rn.
- The elements of IIIB to IIB i.e., from groups 3 to 12 are called as **transition elements**.
- However the Lanthanoids and Actinoids are also considered to be the part of IIIB group (i.e., group 3). These are usually called as **inner transition elements**.
- The elements present in a group show similar physical & chemical properties since they have similar outer electronic configurations

The long form of periodic table superior to Mendeleev's table

- (i) Long form of the periodic table is based upon atomic number which is a more fundamental property of the atom.
- (ii) It relates the position of an element in the periodic table with its electronic configuration.
- (iii) It removes the anomaly about the position of isotopes in the Mendeleev's Table.
- (iv) It removes the anomaly of inversions in the Mendeleev's Table. Thus $^{40}_{18}\text{Ar}$ precedes $^{39}_{19}\text{K}$ in the periodic table because it has a lower atomic number though its atomic mass is more than that of potassium.
- (v) It also removes the other anomalies found in Mendeleev's periodic table (like grouping of chemically dissimilar elements and separation of chemically similar elements etc.).
- (vi) It reflects more clearly the changes in the properties of various elements while moving across a period or down a group.
- (vii) Typical (representative) and transition elements have been separated in this form of the periodic table.
- (viii) It also separates metals and non-metals.

Classification of elements based on electronic configuration

On the basis of electronic configuration, the elements are of four types:

s-Block Elements

General electronic configuration: ns^{1-2} Characteristics of s-block elements:

- (i) All the elements are soft metals.
 - (ii) They have low melting and boiling points.
 - (iii) They are highly reactive.
 - (iv) Most of them impart colours to the flame.
 - (v) They generally form ionic compounds.
 - (vi) They are good conductors of heat and electricity.
- p-Block Elements

General electronic configuration: ns^2np^{1-6}

p-block elements:

- (i) The compounds of these elements are mostly covalent in nature.
- (ii) They show variable oxidation states.
- (iii) In moving from left to right in a period, the non-metallic character of the elements increases.
- (iv) The reactivity of elements in a group generally decreases downwards.
- (v) At the end of each period is a noble gas element with a closed valence shell $ns^2 np^6$ configuration.
- (vi) Metallic character increases as we go down the group.

d-Block Elements

General electronic configuration: $(n-1) d^{1-10} ns^{0-2}$

The d-block elements are known as transition elements because they have incompletely filled d-orbitals in their ground state or in any of the oxidation states.

Characteristics of d-block elements:

- (i) They are all metals with high melting and boiling points.
- (ii) The compounds of the elements are generally paramagnetic in nature.
- (iii) They mostly form coloured ions, exhibit variable valence (oxidation states).
- (iv) They are often used as catalysts.

f-Block Elements

General electronic configuration: $(n - 2) f^{1-14} (n - 1) d^{0-1} ns^2$

They are known as inner transition elements because in the transition elements of d-block, the electrons are filled in $(n - 1) d$ sub-shell while in the inner transition elements of f-block the filling of electrons takes place in $(n - 2) f$ subshell, which happens to be one inner subshell. Characteristics of f-Block elements:

- (i) The two rows of elements at the bottom of the Periodic Table, called the Lanthanoids Ce ($Z = 58$) – Lu ($Z = 71$) and Actinoids Th ($Z = 90$) – Lr ($Z = 103$).
- (ii) These two series of elements are called Inner Transition Elements (f-Block Elements).
- (iii) They are all metals. Within each series, the properties of the elements are quite similar.
- (iv) Most of the elements of the actinoid series are radio-active in nature

Position of Hydrogen

The position of elements on the periodic table largely depends on their electronic configuration. Hydrogen has the electronic configuration of 1. It can get rid of one electron to attain noble gas configuration. This characteristic of hydrogen matches those of alkali metals. But they hydrogen atoms can also gain one electron similar to halogens.

Similarities to Metals

Hydrogen shares many similarities with alkali metals, i.e. elements in group I-A. This is one of the factors that dictates the position of hydrogen in the table. Let us take a look at the similarities

Electronic Configuration: Like all the elements of the group, Hydrogen also has one electron in its last shell, the valence shell. Let us take a look at the composition of valence shells of a few of these alkali metals.

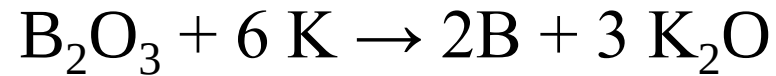
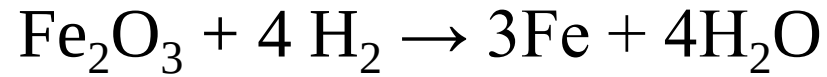
H ($z=1$) : $1s^1$

Li ($z=3$) : $1s^2 2s^1$

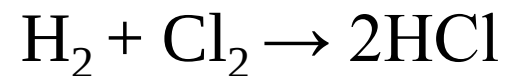
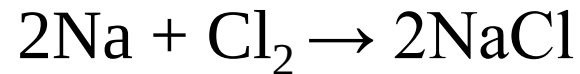
Na ($z=11$) : $1s^2 2s^2 2p^6 3s^1$

Noble Gas Configuration: Like alkali metals, it needs to lose one electron in its valence shell to achieve stable configuration as that of the next noble gas, which in this case is helium. It hence forms the H^+ ion.

Reducing Agent: Hydrogen is a strong reducing agent like all the other alkali metals.



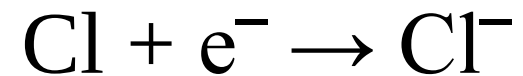
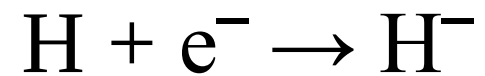
Forms Halides: Also just like alkali metals, hydrogen combines with electronegative elements to form halides



Similarities to Halogens

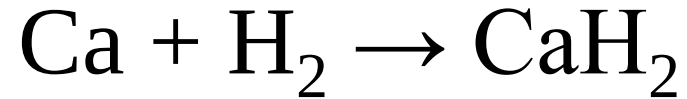
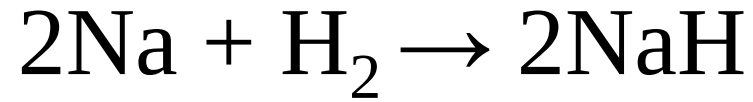
Noble Gas Configuration: Hydrogen can gain one electron to complete its valence shells. Halogens also have seven electrons in their last shell and can gain one electron to gain noble gas configuration.

Electronegativity: They also share the same electronegative nature. Hydrogen also gains one electron (not loses) to become stable and so do halogens.



Diatomic Molecules: Both hydrogen and halogens form diatomic molecules. Hydrogen forms H_2 and, halogens are Cl_2 , F_2 etc.

Reaction with Metals: Hydrogen combines with metals to form metallic hydrides. Similarly, halogens also combine with metals to form metal halides.



Covalent Bonding: Halogens and hydrogen both also combine with non-metals to form molecules with covalent bonding.



Periodicity of properties

Repetition of properties of elements at regular intervals is called periodicity in properties. Some of the properties are discussed below.

- **Atomic and ionic radii**

The size of an atom can be visualized from its atomic radius. Atomic and ionic radius is generally defined as the distance between the centers of the nucleus and the outermost shell of electrons in an atom or ion.

(a) In a Period

Atomic and ionic radii both decrease from left to right across a period in the periodic table because the electrons are added to the orbitals of the same main energy level. The addition of each electron, the nuclear charge (atomic number) increases one by one. The increase in nuclear charge attracts the electrons more strongly close to the nucleus and thus decreases the size of the atoms.

TABLE 3.3. Atomic Radii/pm Across the Periods

Atom (Period II)	Li	Be	B	C	N	O	F
Atomic radius	152	111	88	77	74	66	64
Atom (Period III)	Na	Mg	Al	Si	P	S	Cl
Atomic radius	186	160	143	117	110	104	99

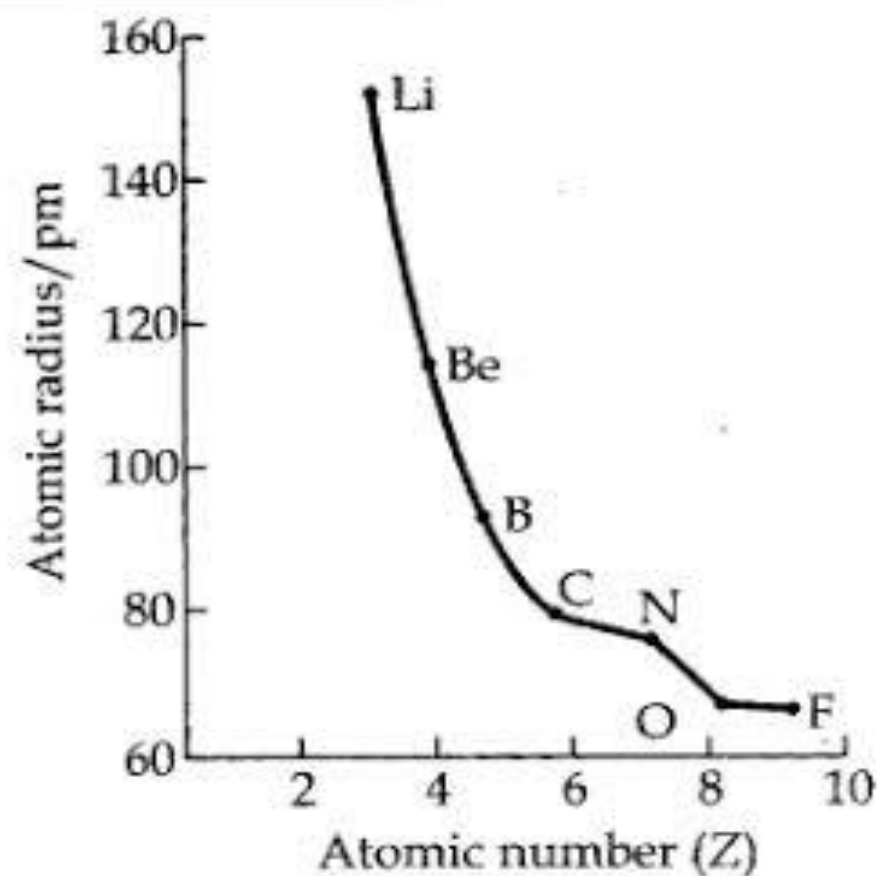


Fig. 3.2 Variation of atomic radius with atomic number across the second period.

(b) In a Group

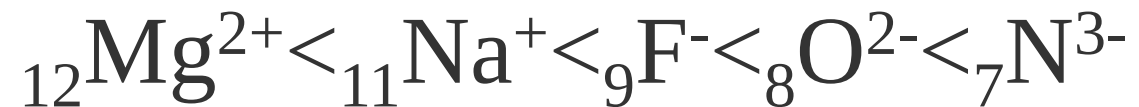
On moving down a group of regular elements both atomic and ionic radii increase because the electrons are added to higher main energy levels, which are on the average, farther from the nucleus. This effect decreases the electrostatic attraction between the nucleus and the valence shell electrons and this decreased electrostatic attraction increases the atomic and ionic radii.

Atom (Group I)	Atomic Radius	Atom (Group 17)	Atomic Radius
Li	152	F	64
Na	186	Cl	99
K	231	Br	114
Rb	244	I	133
Cs	262	At	140

Question:How would you arrange the ions N^{3-} , O^{2-} , Mg^{2+} , Na^{+} , and F^{-} in order of increasing ionic radius?

Answer:If the ions derived from different atoms are **isoelectronic** species, then they all have same number of electrons in their electronic shells and will have **got same electronic configuration** but their nuclear charge will differ because of their difference in number of protons in the nucleus. **With increase in number of protons** in the nucleus the electrons are more attracted towards nucleus thereby causing **the decrease in ionic radius**.

Hence the increasing order of ionic radius is



Question: Why Na^+ is smaller than Na atom?

Answer: In the formation of cation, the radius of the cation is generally smaller than its atom. For Na, when it loses an electron, its outer shell will be removed thus having a smaller ionic radius. It will have a similar configuration to that of Neon. The electronic configuration for Na atom is 2,8,1 ($z=11$) and electronic configuration for Na ion is 2,8 ($z=10$). There is a removal of the m shell and hence, the ion is smaller in size. After removal of one electron, the effective nuclear charge on sodium atoms increase due to which nucleus tends to attract the electrons with a greater force and thus, causing a shrink in size.

Ionisation Energy

The amount of energy required to remove an electron from an isolated gaseous atom in its lowest energy state(ground state) to produce a cation is known as ionisation potential of that element.



Factors affecting ionisation energy

(a) **Nuclear charge:** Ionisation potential increases with the increase in nuclear charge between outermost electrons and nucleus

Element	Proton number	Symbol	First ionisation energy (kJ/mol)
sodium	11	Na	496
magnesium	12	Mg	738
aluminium	13	Al	578
silicon	14	Si	789
phosphorus	15	P	1012
sulphur	16	S	1000
chlorine	17	Cl	1251
argon	18	Ar	1521

(b) Screening effect or Shielding Effect:

Screening effect is also known as the shielding effect. The phenomenon which occurs when the nucleus reduces its force of attraction on the valence electrons due to the presence of electrons in the inner-shell. This is known as a screening effect.

Outer electrons experience attraction from the nucleus and repulsion from the inner electrons. As the attraction between the nucleus decreases, the repulsion increases between the inner electrons and outer electrons.

Ionisation energy decrease with increase in screening power of inner electrons.

(c) Stability of half filled and fully filled orbitals:

According to Hund's Rule atom having half filled or fully filled orbitals are comparatively more stable and hence more energy is needed to remove an electron from such atoms. Half filled p^3, d^5, f^7 or fully filled s^2, p^6, d^{10}, f^{14} are more stable.

Question: Why does Be have higher ionisation energy than B?

Answer: The first ionization of Beryllium is higher than that of Boron because Beryllium has a stable complete electronic configuration ($1s^2 2s^2$) so it, require more energy to remove the first electron from it, where as Boron has electronic configuration ($1s^2 2s^2 2p^1$) which need lesser energy than that of Beryllium.

Electron Affinity

The amount of energy released when an electron is added to a neutral atom in its lowest energy state (ground state) to produce an anion is called its electron affinity.



Electron Affinities (kJ/mol)

1A							8A
H -73							He >0
Li -60	2A Be >0	3A B -27	4A C -122	5A N >0	6A O -141	7A F -328	Ne >0
Na -53	Mg >0	Al -43	Si -134	P -72	S -200	Cl -349	Ar >0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr >0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe >0

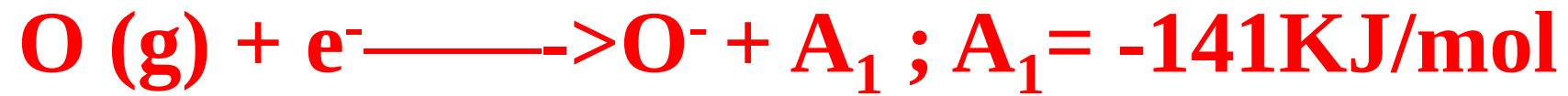
Question: Why Cl has greater electron affinity than F?

Answer: From down a group, electron affinity typically decreases. This is because the atomic radius increases down a group. The electron gained ends up in the outermost shell. It is farther away from the nucleus and thus feels a weaker attraction. Less energy is released.

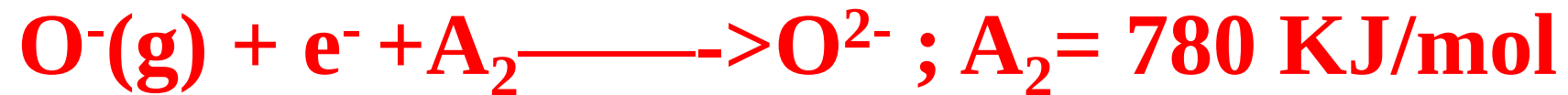
However, there are exceptions. Fluorine, which is higher up the group than chlorine, has a lower electron affinity. This is because the electrons in the outermost shell of a fluorine atom are closer together. The electron gained also feels a great amount of repulsion from the electrons originally in the outermost shell. Energy is required to keep the gained electron in the shell, causing fluorine to have a smaller electron affinity than chlorine.

Question:The 1st electron affinity of oxygen is negative but 2nd electron affinity is positive.

Answer: The 1st electron affinity of oxygen is negative as in the process of adding an electron to the neutral oxygen atom, the energy is released. It is exothermic process.



But during the addition of second electron, energy needs to be put in. Hence, an endothermic reaction takes place.



So in the case of second electron affinity, a negative electron is being added to the negative ion. Repulsion is faced by the electron to be added. To overcome this electrostatic repulsion, the energy needs to be supplied, which makes the 2nd electron affinity positive.

Electronegativity

The tendency of an atom to attract share of electrons towards itself is called electronegativity.

Example: $A-B \longrightarrow A^- + B^+$

In a Period: As we move across a period from left to right the nuclear charge increases and the atomic size decreases, therefore the value of electronegativity increases across a period in the modern periodic table

In a Group:

There is an increase in the atomic number as we move down the group in the modern periodic table. The nuclear charge also increases but the effect of the increase in nuclear charge is overcome by the addition of one shell. Hence, the value of electronegativity decreases as we move down the group.

TABLE 1.1 ELECTRONEGATIVITY VALUES OF SOME COMMON ELEMENTS

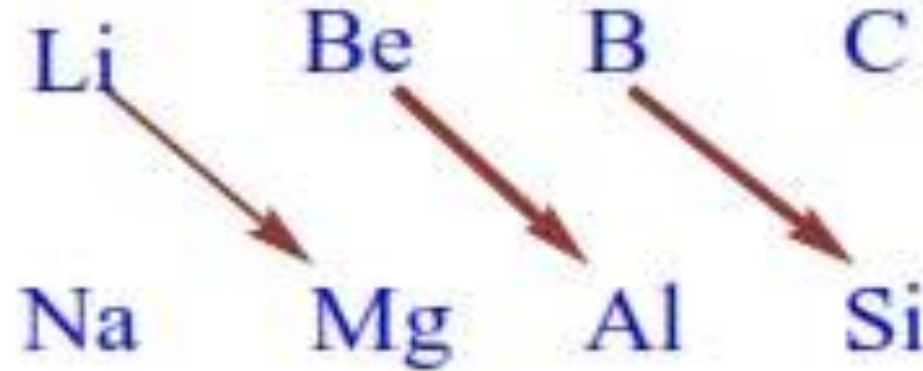
Increasing electronegativity →

			H 2.1				
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	↑ Increasing electronegativity
K 0.8						Br 2.8	

Diagonal Relationship

The similarity in chemical properties between elements in the second period with elements which are lying to their right in the 3rd period or lying diagonally opposite to them.

The diagonal relationship



- Li and N
- Such similarities in chemical properties between the first element and the second element in the next higher group are called diagonal relationship. This kind of relationship exists between some elements of periods 2 and 3.

Similarities between Li and Mg

1. Both form normal oxides when they burn in oxygen.



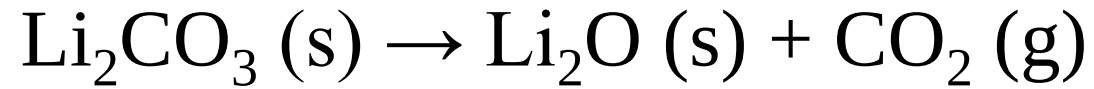
2. They combine with nitrogen to form nitrides.



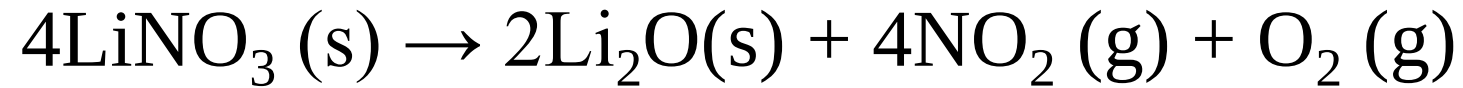
3. They combine with carbon to form carbide



4. Their carbonates, hydroxides and peroxides readily decompose to the oxides on heating.



5. Their nitrates decompose on heating to give oxides, nitrogen dioxide and oxygen.



6. Their carbonates, oxalates and phosphates are sparingly soluble in water.

7. Form strongly hydrated ions in solution

8. Chlorides are deliquescent Chlorides are soluble in ethanol